

BEYOND WORDS

AI POWERED HAND GESTURE RECOGNITION AND CHATBOT FOR ASSISTIVE COMMUNICATION

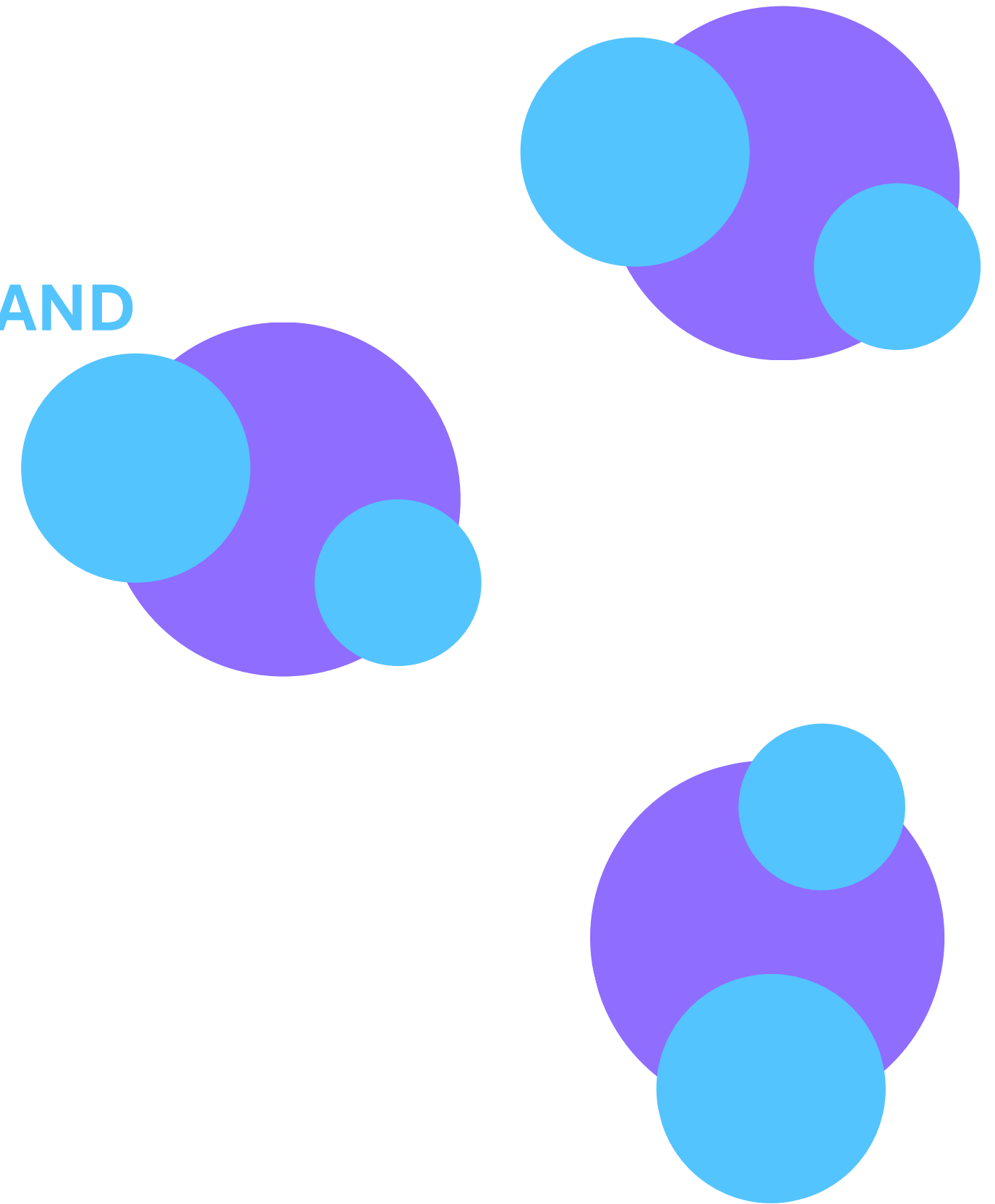
Bridging communication through Artificial Intelligence

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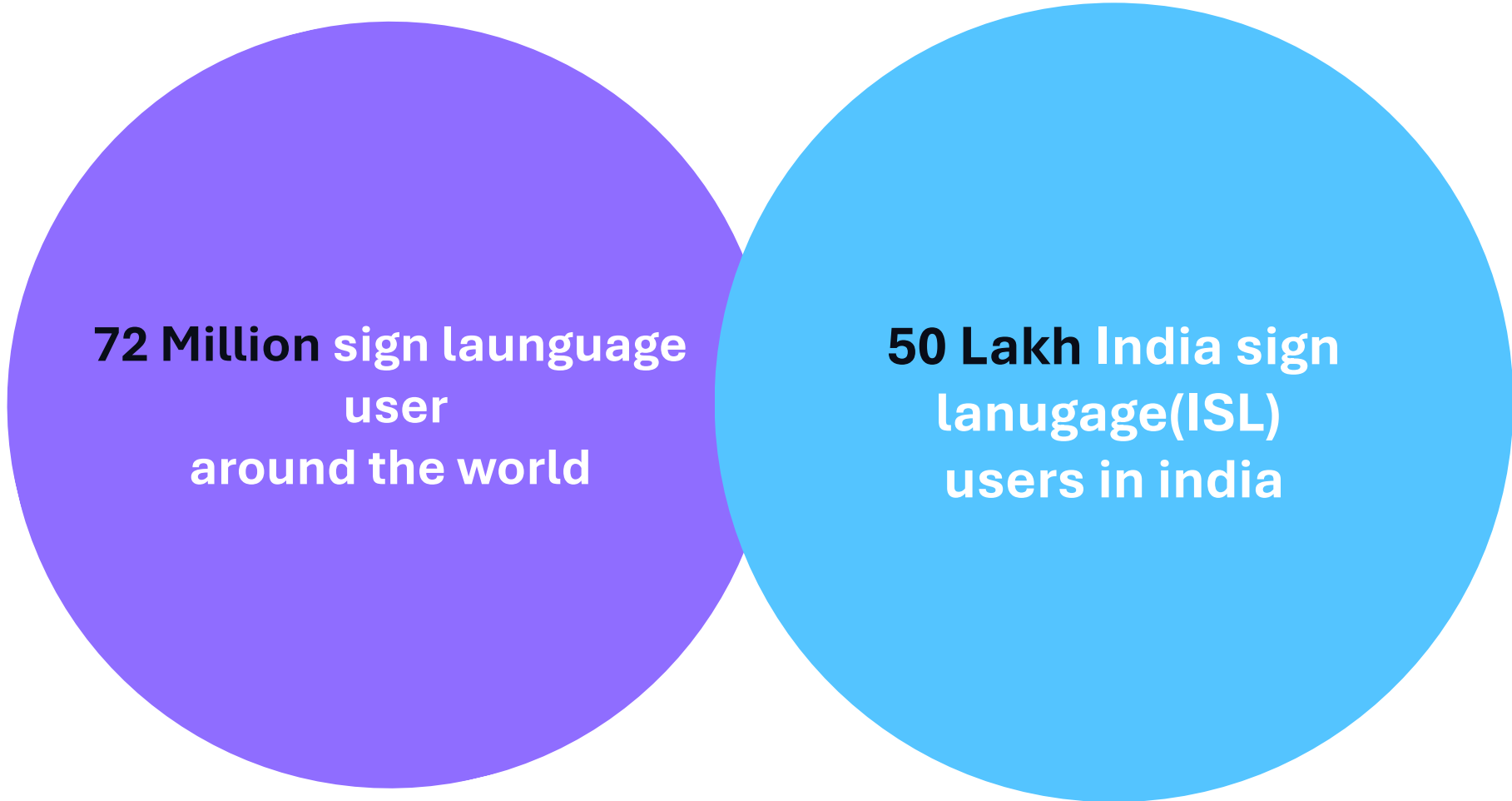
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Problem Statement

Why communication still breaks down

- Sign language is an important form of communication.
 - There are 72 Million sign language user around the world
 - There are 50 Lakh India sign language(ISL) users in india
- Many people do not understand sign language.
- This creates a communication barrier between sign-language users and others.
- Human interpreters are not always available.
- An AI-based solution can help translate signs into understandable communication.



72 Million sign language
user
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Existing System

Current technologies that form the foundation

Human Interpreters

Interpretation often depends on trained human support.

Gesture Recognition

Computer vision can detect and recognize individual gestures.

Machine Learning

Models can classify recognized hand gestures.

Text-to-Speech

Recognized text can be converted into spoken audio.

LIMITATION

Most existing solutions focus on individual stages such as gesture detection, sign recognition, or speech generation rather than providing a single integrated, real-time communication system.

Research Gap

From separate components to one communication pipeline

CURRENT APPROACH

Gesture detection

Sign recognition

Text conversion

Speech generation

GAP

Limited integration

Continuous signs & context

Real-time communication

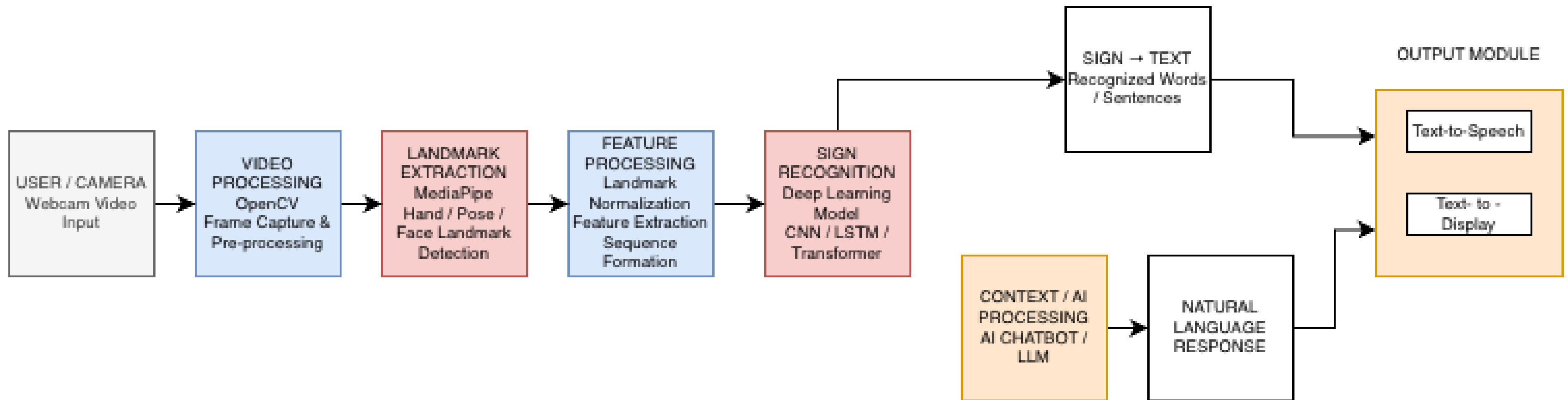
Simple accessibility

OUR FOCUS

Combine recognition, language processing and speech output, intergrate AI Chat bot into one system.

Proposed System

System Architecture :A single pipeline from gesture to voice



Proposed Work

The proposed system integrates computer vision, AI/ML, language processing, text-to-speech, and a chatbot into a single communication platform.

Key Features

- Real-Time Sign Detection
 - Capture hand gestures through a webcam and detect relevant hand movements.
- AI-Based Sign Recognition
 - Use an AI/ML model to identify and interpret recognized sign-language gestures.
- Sign → Text Conversion
 - Convert recognized signs into meaningful text for the non-signing user.
- Text-to-Speech
 - Convert the interpreted text into voice output, enabling easier communication.
- AI Chatbot — Key Feature
 - A built-in chatbot allows users to interact through text or voice, ask questions, continue conversations, and receive AI-generated responses. This extends the system beyond simple one-way sign translation.
- User-Friendly Interface
 - Provide a simple interface combining sign translation, text, speech, and chatbot interaction in one application.

Implementation

Real-Time Gesture Detection

Technology Used :

Python — Core programming language

OpenCV — Captures and processes webcam video

MediaPipe — Used for hand/landmark detection

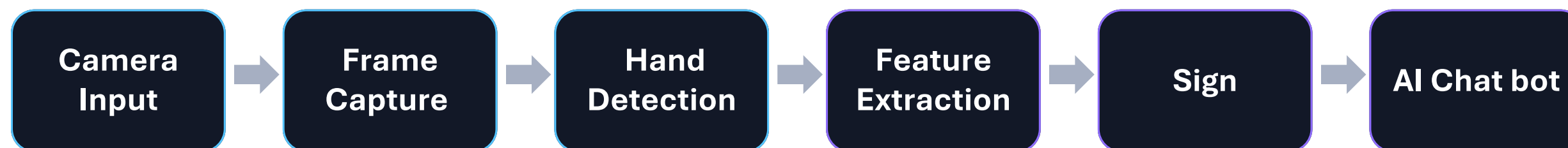
AI/ML Model — Recognizes the detected sign

Text-to-Speech — Converts recognized text into voice output

Current Implementation:

The prototype captures live video through the webcam and processes each frame using computer vision techniques.

Implementation Flow:



```
import cv2.py > ...
1  import cv2
2  import numpy as np
3  import os
4  import time
5  import mediapipe as mp
6  from matplotlib import pyplot as plt
7
8  # Open webcam
9  cap = cv2.VideoCapture(1)
10
11 # Check if camera opened successfully
12 if not cap.isOpened():
13     print("✗ Cannot open camera")
14     exit()
15
16 while True:
17     # Read a frame
18     ret, frame = cap.read()
19
20     # Check if frame was captured
21     if not ret:
22         print("✗ Failed to grab frame")
23         break
24
25     # Display the frame
26     cv2.imshow("OpenCV Feed", frame)
27
28     # Press 'q' to quit
29     if cv2.waitKey(1) & 0xFF == ord('q'):
30         break
31
32 # Release camera and close all windows
33 cap.release()
34 cv2.destroyAllWindows()
```

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